
Homework 2

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MA 564 Section A1

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Problem 1

We use proof by induction. For the case of $n = 1$, we trivially have

$$\bigcup_{i=1}^1 \overline{A_i} = \overline{A_1} = \overline{\bigcup_{i=1}^1 A_i}.$$

I will also show the case for $n = 2$ as a lemma to the inductive step. Let $x \in \overline{A_1 \cup A_2}$. It follows that for all closed sets K containing $A_1 \cup A_2$, we have $x \in K$. Note that since $A_1 \subseteq \overline{A_1}$ and $A_2 \subseteq \overline{A_2}$, we have $A_1 \cup A_2 \subseteq \overline{A_1} \cup \overline{A_2}$ fairly trivially. Since $\overline{A_1}$ and $\overline{A_2}$ are closed, and the union $\overline{A_1} \cup \overline{A_2}$ is finite, we have that $\overline{A_1} \cup \overline{A_2}$ is closed. Therefore, $\overline{A_1} \cup \overline{A_2}$ is a closed set containing $A_1 \cup A_2$, and $x \in \overline{A_1} \cup \overline{A_2}$. Therefore, $\overline{A_1 \cup A_2} \subseteq \overline{A_1} \cup \overline{A_2}$.

Now the other direction. Let $x \in \overline{A_1} \cup \overline{A_2}$. It follows that $x \in \overline{A_1}$ or $x \in \overline{A_2}$. Without loss of generality, assume $x \in \overline{A_1}$. Note that $A_1 \subseteq A_1 \cup A_2 \subseteq \overline{A_1 \cup A_2}$, and thus $x \in \overline{A_1 \cup A_2}$. Therefore, $\overline{A_1} \cup \overline{A_2} \subseteq \overline{A_1 \cup A_2}$.

Now for the inductive case. Let the statement hold for some $n > 1$, and consider $n + 1$. We have

$$\overline{\bigcup_{i=1}^{n+1} A_i} = \overline{\left(\bigcup_{i=1}^n A_i \right) \cup A_{n+1}}.$$

Since $\bigcup_{i=1}^n A_i$ and A_{n+1} are just subsets of X , this is the closure of the union of two subsets of X . By the previous lemma, we have that

$$\overline{\left(\bigcup_{i=1}^n A_i \right) \cup A_{n+1}} = \overline{\left(\bigcup_{i=1}^n A_i \right)} \cup \overline{A_{n+1}}.$$

By the inductive hypothesis, we have

$$\overline{\left(\bigcup_{i=1}^n A_i \right)} \cup \overline{A_{n+1}} = \overline{\left(\bigcup_{i=1}^n \overline{A_i} \right)} \cup \overline{A_{n+1}} = \bigcup_{i=1}^{n+1} \overline{A_i}.$$

By induction, the statement is proven.

For a counterexample, let $X = \mathbb{R}$ with the standard topology, and let $A_i = \left\{ \frac{1}{n} \right\}$ for $n \in \mathbb{N}$. Since $\mathbb{R}$ is Hausdorff, by the previous homework we have that $\{x\}$ is closed for all $x \in \mathbb{R}$, and thus

$$\overline{\left\{ \frac{1}{n} \right\}} = \left\{ \frac{1}{n} \right\}.$$

It follows that

$$\bigcup_{n \in \mathbb{N}} \overline{\left\{ \frac{1}{n} \right\}}.$$

Note that there exists no natural number n such that $1/n = 0$, since this would imply that $1 = 0$, which is clearly a contradiction. I will now show that 0 is a limit point of the sequence $\{1/n\}$, and thus

$$0 \in \overline{\bigcup_{n \in \mathbb{N}} \left\{ \frac{1}{n} \right\}}.$$

Note that we are showing it is a *limit point* of the sequence, and not a point the sequence converges to. These notions are the same in $\mathbb{R}$ I believe, but I don't need to use this fact for the proof. It suffices to show that for all open balls $B_r(0)$, there exists $1/n \in B_r(0)$ for some $n \in \mathbb{N}$, since any open set containing 0 must also contain one of these balls by definition of a basis, and must then contain $1/n$. With this understanding, let $r \in \mathbb{R}^+$, and consider the open neighborhood $B_r(0)$. By the Archimedean property, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < r$. Therefore, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} \in B_r(0) \cup \bigcup_{n \in \mathbb{N}} \left\{ \frac{1}{n} \right\}$$

for all r . Since $1/n \neq 0$ for all $n \in \mathbb{N}$, we have that every open neighborhood of 0 intersects the union at some point other than 0. Therefore, it is a limit point of the neighborhood, and thus

$$0 \in \overline{\bigcup_{n \in \mathbb{N}} \left\{ \frac{1}{n} \right\}}.$$

Since 0 is not in the other union,

$$\bigcup_{n \in \mathbb{N}} \overline{\left\{ \frac{1}{n} \right\}} \neq \overline{\bigcup_{n \in \mathbb{N}} \left\{ \frac{1}{n} \right\}}.$$

Problem 2

- Clearly $\text{Int}(A) \cup \text{Int}(B) \subseteq \text{Int}(A \cup B)$, but the converse does not hold in general. Consider a space X with nonempty subsets A, B, U such that $U \subsetneq A \cup B$, but $U \cap A \neq U$ and $U \cap B \neq U$. That is, U is partially contained by A and B individually, and fully contained by their union. Now consider the topology $\mathcal{T} = \{\emptyset, U, X\}$ on X . Note that this set is closed under union and intersection:

$$\begin{aligned} U \cup \emptyset &= U, & U \cup X &= X, \\ U \cap \emptyset &= \emptyset, & U \cap X &= U. \end{aligned}$$

Therefore, $\mathcal{T}$ is a topology on X . Also note that

$$\text{Int}(A \cup B) = \emptyset \cup U = U,$$

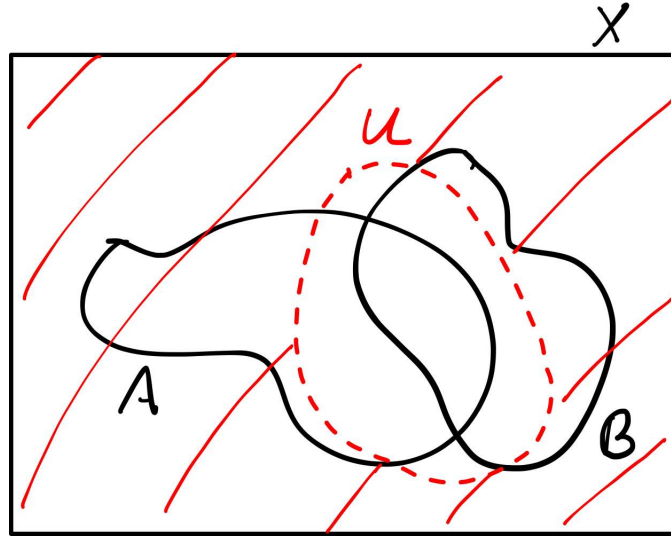
but since U is not a subset of A or B , we have that $\text{Int}(A) = \text{Int}(B) = \emptyset$. Since $\emptyset \cup \emptyset = \emptyset \neq U$, the statement does not hold by counterexample.

- This statement holds. Note that $\text{Int}(A) \subseteq A$ and $\text{Int}(B) \subseteq B$, so $\text{Int}(A) \cap \text{Int}(B) \subseteq A \cap B$. Also note that since $\text{Int}(A), \text{Int}(B)$ are open, this is a finite intersection of open sets, so $\text{Int}(A) \cap \text{Int}(B)$ is open. Since $\text{Int}(A) \cap \text{Int}(B) \subseteq A \cap B$, it follows that $\text{Int}(A) \cap \text{Int}(B) \subseteq \text{Int}(A \cap B)$.

Now the other direction. Let $x \in \text{Int}(A \cap B)$. It follows that there exists some open set $U \subseteq A \cap B$ containing x . It follows that $U \subseteq A$ and $U \subseteq B$, and since U is open, we have $U \subseteq \text{Int}(A), \text{Int}(B)$. Therefore, $U \subseteq \text{Int}(A) \cap \text{Int}(B)$, and thus $x \in \text{Int}(A) \cap \text{Int}(B)$. Therefore, $\text{Int}(A \cap B) \subseteq \text{Int}(A) \cap \text{Int}(B)$, and we are done.

- It's easily shown $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$, but this statement fails in the other direction, by similar logic to the first statement. Let $A, B, U \subsetneq X$ be nonempty sets such that U and $A \cap B$ are disjoint, but U and A are not disjoint, and U and B are not disjoint. Once again $\mathcal{T} = \{\emptyset, U, X\}$ is a topology by the same argument as in statement 1. Consider the closed set U^c , which I will denote V for simplicity. Since U and $A \cap B$ are disjoint, this implies that $A \cap B \subseteq X \setminus U = V$, and thus $V \cap (A \cap B) = A \cap B$. Since V is the only closed set other than $\emptyset$ and X , and $V \subsetneq X$, we have that $\overline{A \cap B} = V$, since this is the smallest closed set containing $A \cap B$. However, since U and A are not disjoint, we have that $U \cap A \neq \emptyset$, and V does not contain $U \cap A$. Therefore, $V \cap A \neq A$, and thus A is not a subset of V . By the same logic, B is not a subset of V either. Therefore, $\overline{A} = X$ and $\overline{B} = X$, and since $X \cap X = X \neq V$, we have that $\overline{A \cap B} \neq \overline{A} \cap \overline{B}$.

Here's a helpful image to visualize the argument I'm making.



Problem 3

Let A be closed. Then $A = \overline{A}$. Note that $\overline{A} \setminus \text{Int}(A) \subseteq \overline{A}$, and $\partial A = \overline{A} \setminus \text{Int}(A)$. Therefore, $\partial A \subseteq \overline{A} = A$.

Now for the converse, let $\partial A \subseteq A$. It follows that $\overline{A} \setminus \text{Int}(A) \subseteq A$. Note that $\overline{A} = \text{Int}(A) \cup (\overline{A} \setminus \text{Int}(A))$. Since $\text{Int}(A) \subseteq A$ and $\overline{A} \setminus \text{Int}(A) = \partial A \subseteq A$, we have that their union must also be a subset of A . That is,

$$\overline{A} = \text{Int}(A) \cup (\overline{A} \setminus \text{Int}(A)) \subseteq A.$$

Since we also know for all sets that $A \subseteq \overline{A}$, we have that $A = \overline{A}$, and A is closed.

Problem 4

First we prove a brief lemma: for any function $f : X \rightarrow X$ for X any set, and for any collection $\{U_i\}_{i \in I}$ of subsets of X ,

$$\bigcup_{i \in I} f^{-1}(U_i) = f^{-1}\left(\bigcup_{i \in I} U_i\right).$$

Let $x \in \bigcup_{i \in I} f^{-1}(U_i)$. It follows that there exists $i \in I$ such that $x \in f^{-1}(U_i)$, and thus $f(x) \in U_i$. Since $U_i \subseteq \bigcup_{i \in I} U_i$, we have that $f(x) \in \bigcup_{i \in I} U_i$, and thus

$$x \in f^{-1} \left(\bigcup_{i \in I} U_i \right).$$

Therefore, $\bigcup_{i \in I} f^{-1}(U_i) \subseteq f^{-1} \left(\bigcup_{i \in I} U_i \right)$. Now for the converse, let $x \in f^{-1} \left(\bigcup_{i \in I} U_i \right)$. It follows that $f(x) \in \bigcup_{i \in I} U_i$, and thus there exists $i \in I$ such that $f(x) \in U_i$. Therefore, $x \in f^{-1}(U_i)$, and since $f^{-1}(U_i) \subseteq \bigcup_{i \in I} f^{-1}(U_i)$, we have

$$x \in \bigcup_{i \in I} f^{-1}(U_i).$$

This proves the lemma. Since every open set in $\mathbb{R}$ can be represented as a union of basis elements, it suffices to show the preimage of a basis element is open, since then the preimage of the union would become a union of open sets, and thus open.

Let $(f \times g) : \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R}$ be the map $x \mapsto (f(x), g(x))$. Let U, V be open in $\mathbb{R}$. Since f and g are continuous, $f^{-1}(U)$ and $g^{-1}(V)$ are open in $\mathbb{R}$. Note that $(f \times g)^{-1}(U \times V)$ is simply the set of all $x \in \mathbb{R}$ such that $f(x) \in U$ and $g(x) \in V$. That is,

$$(f \times g)^{-1}(U \times V) = f^{-1}(U) \cap g^{-1}(V).$$

Since this is an intersection of open sets, $(f \times g)^{-1}(U \times V)$ is open. Since $U \times V$ is a basis element for the product topology on $\mathbb{R} \times \mathbb{R}$, this suffices to show $(f \times g)$ is continuous.

Now let $h : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be the addition map $(x, y) \mapsto x + y$. Let $B_r(z)$ be an open set in $\mathbb{R}$. Note that the preimage

$$h^{-1}(B_r(z)) = \{(x, y) \in \mathbb{R}^2 \mid d(x + y, z) < r\}.$$

We wish to show this set is open. It suffices to show we can form an open ball about any point in $h^{-1}(B_r(z))$ which is a subset of $h^{-1}(B_r(z))$, since we could then take the union over all such sets, as the union of open sets is open. Let $(x, y) \in B_r(z)$, and let

$$s := \frac{r - d(x + y, z)}{2}$$

for simplicity. Consider the open set $B_s(x) \times B_s(y)$. Let $(a, b) \in B_s(x) \times B_s(y)$. It follows that

$$d(x, a), d(y, b) < s.$$

Equivalently,

$$|x - a| + |y - b| < 2s.$$

By the triangle inequality,

$$|x + y - a - b| \leq |x - a| + |y - b| < 2s.$$

Therefore, $d(x + y, a + b) < 2s$. By the triangle inequality once again,

$$d(a + b, z) \leq d(a + b, x + y) + d(x + y, z) < 2s + d(x + y, z) = r - d(x + y, z) + d(x + y, z) = r.$$

Therefore, $d(a + b, z) < r$, and thus $B_s(x) \times B_s(y) \subseteq h^{-1}(B_r(z))$. By the previous logic, it follows that h is continuous.

Note that

$$(h \circ (f \times g))(x) = h(f(x), g(x)) = f(x) + g(x) = (f + g)(x).$$

We showed in class that the composition of continuous functions is continuous. Therefore, $(f + g)$ is continuous.

Now consider the function $k : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $(x, y) \mapsto (x, -y)$. Note that this function is fairly trivially continuous. Let $U \times V$ be open in $\mathbb{R}^2$. Then the preimage is $U \times (-V)$, where $-V := \{-x \mid x \in V\}$. Since V is open, for all x there exists a radius $r \in \mathbb{R}^+$ such that $B_r(x) \subseteq V$. It follows basically immediately that $B_r(-x) \subseteq -V$:

$$y \in B_r(-x) \implies |y + x| < r \implies |-y - x| < r \implies -y \in B_r(x) \subseteq V \implies y \in -V.$$

Therefore, $-V$ is open in $\mathbb{R}$ by previous logic. Also by previous logic, k is continuous, since all basis sets have open preimages. Note that

$$(h \circ k \circ (f \times g))(x) = (h \circ k)(f(x), g(x)) = h(f(x), -g(x)) = f(x) - g(x) = (f - g)(x).$$

Once again, the composition of continuous functions is continuous, so $(f - g)(x)$ is continuous.

Now let $m : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the multiplication map $(x, y) \mapsto xy$. Let $B_r(z)$ be an open ball in $\mathbb{R}$. We will show m is continuous in the same way as previously; that is, we will show for all elements in the preimage, there exists a basis element containing it which is a proper subset of the preimage. For simplicity, let

$$s := \frac{-(|x| + |y|) + \sqrt{(|x|^2 + |y|^2 + |2xy|) + 4(r - |xy - z|)}}{2}.$$

It follows that s satisfies $s^2 + (|x| + |y|)s - (r - |xy - z|) = 0$. It also follows that s is positive, because since $|xy - z| = d(xy, z) < r$, we have that $4(r - |xy - z|) > 0$, and thus the inside of the square root is greater than the square of $|x| + |y|$. Therefore, we can define the basis element of $\mathbb{R}^2$

$$B_s(x) \times B_s(y).$$

Let $(a, b) \in B_s(x) \times B_s(y)$. It follows that $|a - x|, |b - y| < s$, and thus

$$\begin{aligned} s^2 + (|x| + |y|)s &= r - d(xy, z) \\ \implies |a - x||b - y| + |y||a - x| + |x||b - y| &< r - d(xy, z). \end{aligned}$$

By the triangle inequality and the fact that $|ab| = |a||b|$,

$$\begin{aligned} |a - x||b - y| + |y||a - x| + |x||b - y| &\geq |(a - x)(b - y) + x(b - y) + y(a - x)| \\ &= |ab + xy - ay - bx + bx - xy + ay - xy| \\ &= |ab - xy| \\ &= d(ab, xy). \end{aligned}$$

Therefore, $d(ab, xy) < r - d(xy, z)$. By the triangle inequality, we have

$$d(ab, z) \leq d(ab, xy) + d(xy, z) < r - d(xy, z) + d(xy, z) = r.$$

Therefore, $ab \in B_r(z)$, and thus $(a, b) \in m^{-1}(B_r(z))$. Therefore,

$$B_s(x) \times B_s(y) \subseteq m^{-1}(B_r(z)).$$

Thus, m is continuous. We have that

$$(m \circ (f \times g))(x) = m(f(x), g(x)) = f(x)g(x) = (fg)(x).$$

Since composition of continuous functions is continuous, we have that fg is continuous.

Finally, suppose that $g(x) \neq 0$ for all $x \in \mathbb{R}$. Consider instead g to be a map $\mathbb{R} \rightarrow g(\mathbb{R})$. This is trivially still continuous, since g is continuous. Now consider the map $\ell : \mathbb{R} \times g(\mathbb{R}) \rightarrow \mathbb{R} \times \mathbb{R}$ defined by

$$(x, g(y)) \mapsto \left(x, \frac{1}{g(y)}\right).$$

It should be clear that this is continuous if and only if $g(y) \mapsto \frac{1}{g(y)}$ is continuous, since we have shown the identity map is continuous. Let $B_r(z)$ be open in $\mathbb{R}$, and let $x \in \ell^{-1}(B_r(z))$. I am abusing notation here because I don't feel like defining another function. Note that $x \in \ell^{-1}(B_r(z))$ if and only if $x \in g(\mathbb{R})$ by definition of ℓ . We wish to show we can construct an open ball $B_s(x)$ such that $B_s(x) \cap g(\mathbb{R})$ is a subset of $\ell^{-1}(B_r(z))$. I ran out of time to show this unfortunately because the homework is due in 10 minutes, so I will just assume it.

Finally, we have

$$(m \circ \ell \circ (f \times g)) = (m \circ \ell)(f(x), g(x)) = m\left(f(x), \frac{1}{g(x)}\right) = \frac{f(x)}{g(x)} = \left(\frac{f}{g}\right)(x).$$

Since composition of continuous functions is continuous, f/g is continuous.

Problem 5

We prove that $\text{Graph}(f)^c$ is open, which is equivalent by definition of a closed set. For simplicity, let

$$K := \text{Graph}(f)^c = \{(x, y) \in X \times Y \mid y \neq f(x)\}.$$

Let $(x, y) \in K$. Since Y is Hausdorff, there exist disjoint neighborhoods $N_y, N_{f(x)}$ of y and $f(x)$, respectively. Since f is continuous, $f^{-1}(N_{f(x)})$ is open in X . Note that if $f^{-1}(y) \cap f^{-1}(N_{f(x)}) \neq \emptyset$, then there exists at least one $y' \in f^{-1}(N_{f(x)})$ such that $f(y') = y$. It follows that $y \in N_{f(x)}$, a contradiction. Therefore, $f^{-1}(y)$ and $f^{-1}(N_{f(x)})$ are disjoint. Note that

$$f^{-1}(N_{f(x)}) \times N_y$$

is open in $X \times Y$ since $f^{-1}(N_{f(x)})$ is open in X and N_y is open in Y . Now let $(a, b) \in f^{-1}(N_{f(x)}) \times N_y$. Note that $(a, b) \in \text{Graph}(f)$ if and only if $b = f(a)$. However, $f(a) \in N_{f(x)}$, and $b \in N_y$. Since these sets are disjoint, we have that $b \neq f(a)$, and thus $(a, b) \in K$. Therefore, we have

$$f^{-1}(N_{f(x)}) \times N_y \subseteq K$$

for all $(x, y) \in K$. Now consider the union

$$\bigcup_{(x,y) \in K} f^{-1}(N_{f(x)}) \times N_y,$$

where once again we take $N_{f(x)}$ and N_y to be disjoint open neighborhoods of $f(x)$ and y , which exist since Y is Hausdorff. Note that since each set in the union is a subset of K , we have

$$\bigcup_{(x,y) \in K} f^{-1}(N_{f(x)}) \times N_y \subseteq K.$$

However, since $x \in f^{-1}(N_{f(x)})$ because $f(x) \in N_{f(x)}$, and since $y \in N_y$, we have $(x, y) \in f^{-1}(N_{f(x)}) \times N_y$. Therefore, for all $(x, y) \in K$, there exists a set in the union containing (x, y) . Therefore,

$$K \subseteq \bigcup_{(x,y) \in K} f^{-1}(N_{f(x)}) \times N_y.$$

It follows that K equals this union. Since this is a union of open sets, K is open, and thus $K^c = \text{Graph}(f)$ is closed.

Problem 6

For any subset $U \subseteq Y$, the preimage $f^{-1}(U) \subseteq X$ is a subset of X by definition of f . Since X has the discrete topology, all subsets of X are open, and thus $f^{-1}(U)$ is open. Therefore, f is continuous.

Let $(0, 1) \in K$ be an open set in $X \times Y$. Since K is open, there exist a collection of open sets $\{U_i\}_{i \in I}$ in X , and a corresponding collection of open sets $\{V_i\}_{i \in I}$ in Y , such that

$$K = \bigcup_{i \in I} U_i \times V_i.$$

Since $(0, 1) \in K$, there must exist a set $U_i \times V_i$ containing $(0, 1)$ for some $i \in I$. It follows that V_i must contain 1. However, V_i is also open in Y , and the only open set containing 1 in Y is Y . Therefore, $V_i = Y$, and thus $0 \in V_i$. Since $0 \in U_i$ as well, we have $(0, 0) \in U_i \times V_i$, and thus $(0, 0) \in K$. Therefore, any neighborhood of $(0, 1)$ intersects $\text{Graph}(f)$ at $(0, 0)$. Since $(0, 0) \neq (0, 1)$, it follows that $(0, 1)$ is a limit point of $\text{Graph}(f)$, and thus is contained within the closure of $\text{Graph}(f)$. However, since $f(0) = 0 \neq 1$, we have $(0, 1) \notin \text{Graph}(f)$, and thus we have the proper subset $\text{Graph}(f) \subsetneq \overline{\text{Graph}(f)}$. Therefore, $\text{Graph}(f)$ is not closed.

Problem 7

Let $\pi : \text{Graph}(f) \rightarrow X$ be the canonical projection $(x, f(x)) \mapsto x$, and consider the function $\pi^{-1} : X \rightarrow \text{Graph}(f)$ defined by $x \mapsto (x, f(x))$. Note that this map is well-defined because f is well defined; that is, if $x = y$, then we must have $f(x) = f(y)$ because f is a function, and thus $\pi^{-1}(x) = (x, f(x)) = (y, f(y)) = \pi^{-1}(y)$. Also note that this is clearly an inverse to π :

$$(\pi^{-1}\pi)(x, f(x)) = \pi^{-1}(x) = (x, f(x));$$

$$(\pi\pi^{-1})(x) = \pi(x, f(x)) = x.$$

Therefore, π is a bijection. Now we show π is a homeomorphism. First, we show π is continuous. Let $U \subseteq X$ be open in X . Note that

$$\pi^{-1}(U) = \{(x, f(x)) \in \text{Graph}(f) \mid x \in U\}.$$

I claim that $\pi^{-1}(U) = \text{Graph}(f) \cap (U \times Y)$. Note that since $\text{Graph}(f)$ inherits the subspace topology from the product topology on $X \times Y$, this is an open set. We have that $V_x \times Y$ is open in $X \times Y$, since V is open in X and Y is open in Y . It follows that $\text{Graph}(f) \cap (U \times Y)$ is open in $\text{Graph}(f)$ by definition of the subspace topology.

Now to show these sets are equal. Let $(x, f(x)) \in \pi^{-1}(U)$. It follows that $x \in U$, and since $f(x) \in Y$, we have $(x, f(x)) \in U \times Y$. Since we also have $(x, f(x)) \in \text{Graph}(f)$, we have $(x, f(x)) \in \text{Graph}(f) \cap (U \times Y)$. Therefore, $\pi^{-1}(U) \subseteq \text{Graph}(f) \cap (U \times Y)$.

Now the converse. Let $(x, f(x)) \in \text{Graph}(f) \cap (U \times Y)$. It follows that $x \in U$, and thus $\pi(x, f(x)) = x \in U$. Therefore, $(x, f(x)) \in \pi^{-1}(U)$. Therefore, $\text{Graph}(f) \cap (U \times Y) = \pi^{-1}(U)$.

Now, we show π^{-1} is continuous. Note that $(\pi^{-1})^{-1} = \pi$, so we must show $\pi(U)$ is open for all open sets U . Let $U \subseteq \text{Graph}(f)$ be open. Note that since $\text{Graph}(f)$ is imbued with the subspace topology inherited from the product topology on $X \times Y$. That is, every open set U can be represented as a union

$$\bigcup_{i \in I} \text{Graph}(f) \cap (V_i \times W_i),$$

for $\{V_i\}_{i \in I}$ open in X and $\{W_i\}_{i \in I}$ open in Y . By the same proof given in problem 4,

$$\pi \left(\bigcup_{i \in I} \text{Graph}(f) \cap (V_i \times W_i) \right) = \bigcup_{i \in I} \pi(\text{Graph}(f) \cap (V_i \times W_i)).$$

It then suffices to show $\pi(\text{Graph}(f) \cap (V_i \times W_i))$ is open in X for some fixed $i \in I$. Let $A := \text{Graph}(f) \cap (V \times W)$ for V, W open in X, Y , respectively. First, I will show that $\pi(A) = f^{-1}(W) \cap V$. Let $x \in \pi(A)$. It follows that $x \in V$ and $f(x) \in W$, and thus $x \in f^{-1}(W) \cap V$. Now let $x \in f^{-1}(W) \cap V$. It follows that $x \in V$ and $f(x) \in W$, and thus $(x, f(x)) \in V \times W$. Therefore, $(x, f(x)) \in A$, and $\pi(x, f(x)) = x \in \pi(A)$. Therefore, $\pi(A) = f^{-1}(W) \cap V$. Since f is continuous, $f^{-1}(W)$ is open in X , and by definition so is V . This is then a finite intersection of open sets, which is open. Therefore, $\pi(A)$ is open, and π^{-1} is continuous. Thus, π is a homeomorphism from $\text{Graph}(f)$ to X .